



**R V COLLEGE OF ENGINEERING**  
(An autonomous institution affiliated to VTU, Belgaum)  
DEPARTMENT OF  
**INDUSTRIAL ENGINEERING & MANAGEMENT**

**IM233AI – WORK SYSTEMS DESIGN**  
**UNIT IV**

| Q No | Question                                                                        | Marks | Difficulty Level | BT Level | COs |
|------|---------------------------------------------------------------------------------|-------|------------------|----------|-----|
| 1.   | What is the Japanese term for Waste in Lean Production? Define it.              | 2     | L                | 2        | 2   |
| 2.   | Name the two most detrimental forms of waste identified in Lean production.     | 2     | L                | 1        | 2   |
| 3.   | Why is Overproduction considered the worst form of waste?                       | 2     | M                | 2        | 2   |
| 4.   | Define the Lean waste known as Defects (or Production of defective parts).      | 2     | L                | 1        | 2   |
| 5.   | Define the Lean waste known as Waiting.                                         | 2     | L                | 1        | 2   |
| 6.   | List two "other forms of waste" (non-Muda wastes) often discussed in Lean.      | 2     | L                | 1        | 2   |
| 7.   | How does the Lean concept of Elimination of Waste contribute to cost reduction? | 2     | M                | 2        | 2   |
| 8.   | Define the core principle of Just-in-Time (JIT) Production.                     | 2     | L                | 1        | 2   |
| 9.   | What is a Pull System of Production Control?                                    | 2     | L                | 1        | 2   |
| 10.  | What is the role of Kanban in JIT production?                                   | 2     | M                | 1        | 2   |
| 11.  | Distinguish between an Internal Setup and an External Setup.                    | 2     | M                | 2        | 2   |
| 12.  | What is the Japanese term for Autonomation? Define its core concept.            | 2     | L                | 1        | 2   |
| 13.  | State the two essential elements of Jidoka.                                     | 2     | L                | 1        | 2   |
| 14.  | What does Total Productive Maintenance (TPM) focus on?                          | 2     | L                | 1        | 2   |
| 15.  | Define the Japanese term for Continuous Improvement.                            | 2     | L                | 1        | 2   |
| 16.  | List the five Ss in the 5S methodology (in English).                            | 2     | L                | 1        | 2   |
| 17.  | Define Takt Time and explain its significance in Lean.                          | 2     | M                | 2        | 2   |